



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

IoT-Based Fan Automation with Speed Control

TEAM MEMBERS:

RITIK RAJ (24MCA0261)

ARVIND KUMAR (24MCA0188)

KSHITIZ SHARMA (24MCA0238)

KOLI RAVINDRA HITESH (24MCA0240)

SINGH PRAKASH UDAYA RAJ (24MCA0234)

SUBMITTED TO:

Sir BALAJI E

Abstract

This project introduces an **IoT-based fan automation system** that dynamically adjusts fan speed based on real-time temperature readings. Utilizing **Arduino Uno**, a **DHT11 sensor** and displaying it on **I2C Interface**, the system ensures energy efficiency and remote control. The fan speed is adjusted through a **PWM motor driver(L298N)**, eliminating manual intervention while providing IoT-enabled monitoring via a laptop. The integration of these components provides a foundation for smart home applications, enhancing comfort while optimizing energy usage.

Research Gap

While several fan control systems exist, most rely on **manual operation** or **basic automation without adaptive response to environmental changes**.

Conventional systems lack real-time monitoring, energy optimization, and smart control integration, making them inefficient for modern smart homes.

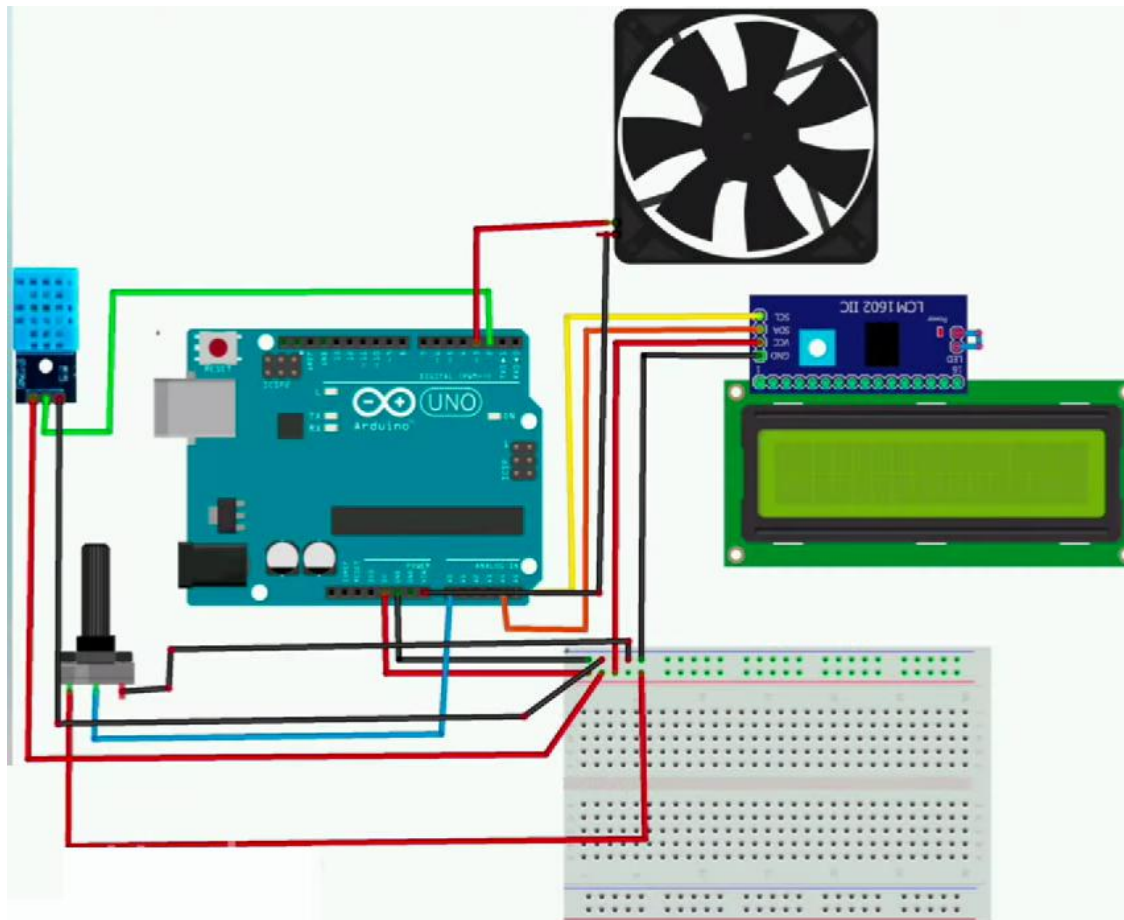
- Moreover, existing temperature-based fan systems often do not:
- Incorporate **IoT features** for intelligent monitoring or control.
- Use cost-effective and **easily replicable components** like the Arduino, DHT11, and L298N.
- Provide **real-time feedback** to the user through a simple user interface such as an I2C LCD.
- Address **energy conservation** through dynamic speed adjustment using temperature thresholds.
- This project fills the gap by creating a **low-cost, energy-efficient, and automated fan control system** with potential for IoT integration—bridging the divide between traditional fan systems and modern smart home automation.

Objective

The objective of this project is to design and implement an IoT-based smart fan automation system that:

- ✓ **Monitors ambient temperature** in real-time using a DHT11 sensor.
- ✓ **Automatically controls the speed of a DC fan** based on temperature thresholds using an L298N motor driver and PWM signals.
- ✓ **Displays live temperature readings** on a 16x2 LCD via I2C interface for user awareness.
- ✓ **Eliminates the need for manual fan operation**, promoting energy efficiency and convenience.
- ✓ **Demonstrates the practical application of Arduino** in smart home automation and IoT systems.

Block Diagram



Explanation:

- **DHT11 Sensor:** Continuously sends temperature data to the Arduino.
- **Arduino Uno:** The brain of the system. Reads temperature, processes logic, controls display and fan speed.
- **16x2 LCD with I2C:** Displays live temperature readings.
- **L298N Motor Driver:** Receives PWM signals from Arduino to adjust fan speed.
- **DC Motor Fan:** Speed varies with temperature (e.g., low/high speed or off).
- **9V Battery:** Powers the whole system.

Working Principle

The system operates on the principle of **temperature-based automation** using a feedback control mechanism. The DHT11 sensor continuously measures the ambient temperature and sends the data to the Arduino Uno. Based on predefined temperature thresholds, the Arduino decides whether the fan should be OFF, run at low speed, or at high speed.

The fan speed is controlled via **Pulse Width Modulation (PWM)** signals sent from the Arduino to the **L298N motor driver**, which then powers the DC motor accordingly. The current temperature is also displayed on a 16x2 LCD using an I2C interface for real-time monitoring.

This system removes the need for manual control and enables the fan to function based on real-time temperature, enhancing energy efficiency and comfort.

Working of the System

1. Temperature Sensing:

The DHT11 sensor detects the current temperature of the surrounding environment.

2. Data Processing:

The Arduino receives this data and compares it with predefined thresholds (e.g., 25°C and 30°C).

3. Fan Control Logic:

- If temperature < 25°C → Fan remains OFF
- If 25°C ≤ temperature < 30°C → Fan runs at low speed
- If temperature ≥ 30°C → Fan runs at high speed

4. **Fan Speed Regulation:**

PWM signals are generated by the Arduino and sent to the L298N motor driver, which adjusts the speed of the fan accordingly.

5. **Display Output:**

The real-time temperature is shown on a 16x2 LCD via I2C, providing the user with immediate feedback.

6. **Power Supply:**

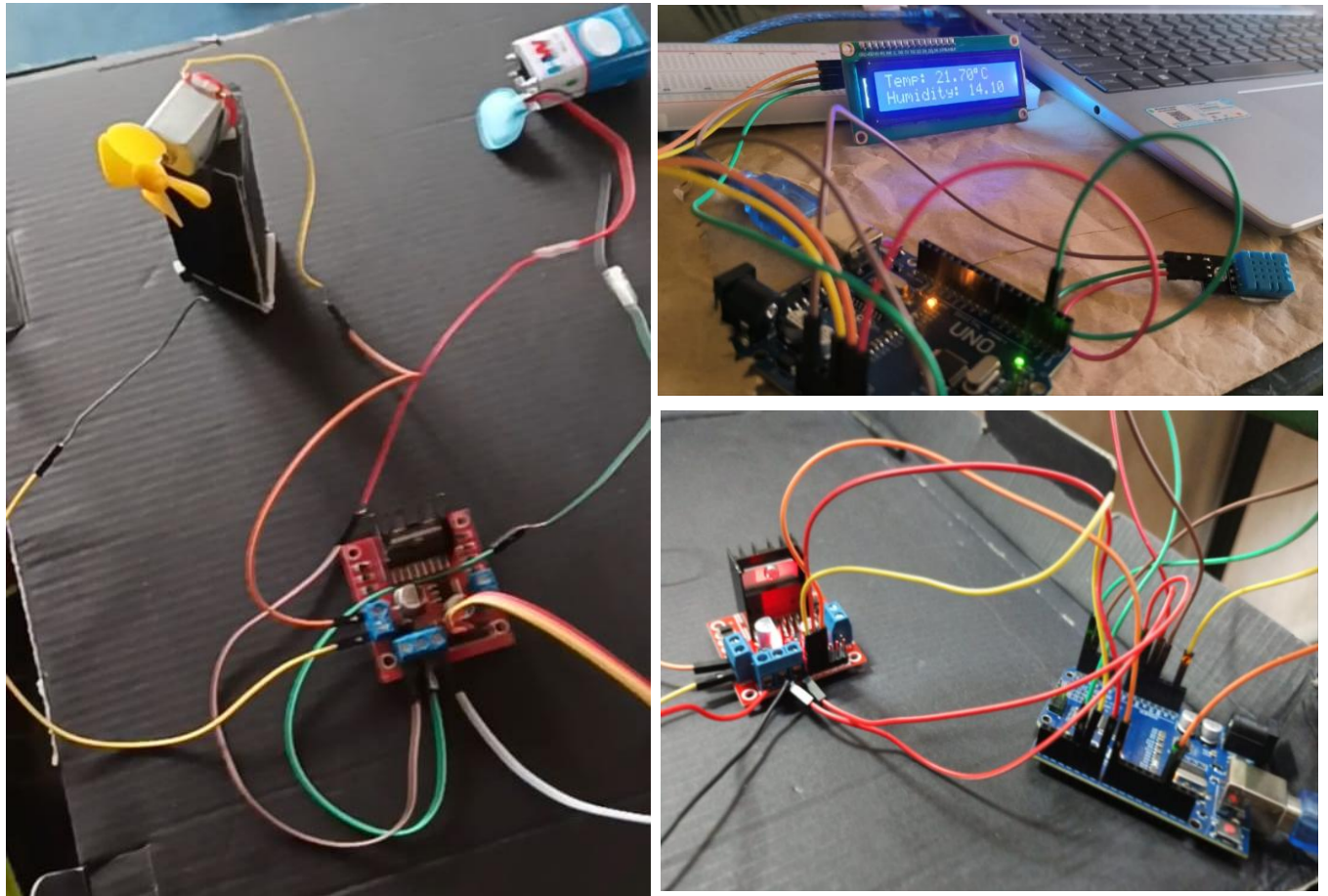
The entire setup is powered by a 9V battery, making it portable and self-sustained.

Results

The IoT-based fan automation system was successfully designed, assembled, and tested. The system operated as expected under various temperature conditions. Below are the observed results:

- When the ambient temperature was **below 25°C**, the fan remained **OFF**, conserving energy.
- When the temperature rose to **between 25°C and 30°C**, the fan turned **ON at low speed**.
- When the temperature exceeded **30°C**, the fan operated at **high speed**.
- The **LCD display** accurately showed the current temperature in real-time, providing clear feedback to the user.
- The **response time** from sensing to fan activation was fast, ensuring real-time control.
- The system proved to be **stable, energy-efficient**, and suitable for basic smart home automation.

Output



Conclusion

The IoT-based smart fan automation system developed in this project successfully demonstrates how simple electronic components can be integrated to create an efficient, temperature-responsive fan control system. By utilizing the Arduino Uno, DHT11 sensor, L298N motor driver, and a DC fan, the system operates autonomously based on real-time environmental conditions without requiring manual intervention.

The system provides:

- **Accurate temperature monitoring**
- **Automatic fan speed control**
- **Real-time display output**

This project lays the foundation for more advanced smart home systems and showcases the practical application of IoT concepts in real-world scenarios.